

Generative AI for Personalized Multisensory Immersive Experiences: Challenges and Opportunities for Stress Reduction

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Figure 1: A 360° tropical beach environment generated using Blockade Labs' skybox tools.



Figure 2: A 360° winter log cabin environment created with Blockade Labs' skybox tools.

ABSTRACT

Stress management and relaxation are critical areas of interest in mental health and well-being. Forest bathing is a practice that has been shown to have a positive effect on reducing stress by stimulating all the senses in an immersive nature experience. Since access to nature is not universally available to everyone, virtual reality has emerged as a promising tool to simulate this type of experience. Furthermore, generative artificial intelligence (GenAI) tools offer new opportunities to create highly personalized and immersive experiences that can enhance relaxation and reduce stress. This study explores the potential of personalized multisensory VR environments, designed using GenAI tools, to optimize relaxation and stress relief via two experiments that are currently underway. The first evaluates the effectiveness of non-personalized versus personalized VR scenes generated using AI tools to promote increased relaxation. The second explores the potential benefits of providing the user with additional personalization tools, from adding new virtual elements to the AI-generated scene, to adding AI-generated sounds and scent/haptics customization. Ultimately, this research aims to identify which customizable elements may lead to improved therapeutic benefits for multisensory VR experiences.

Index Terms: Generative AI, environment personalization, multisensory virtual reality, forest bathing, relaxation.

1 INTRODUCTION

Chronic stress is increasingly recognized as a global health concern with wide-ranging implications for mental health, cognition, and physiological stability. Typical interventions can include pharmacological and non-pharmacological therapies, such as cognitive behavioral therapy [14], mindfulness [16], or yoga [34]. Nature exposure, or the practice of ‘forest bathing’ (shinrin-yoku) [11, 32] is one known relaxation technique that involves immersive, multisensory interaction with nature. Forest bathing has been widely recog-

nized for its ability to reduce stress, alleviate anxiety, and promote relaxation [26], as well as reduce cortisol levels [31], heart rate, and blood pressure [15], and improve mood and relaxation [26, 27].

Despite its effectiveness, access to natural environments is not universally available. Individuals with limited mobility, those confined to hospitals, or people residing in highly urbanized areas often face barriers to experiencing the benefits of nature [8]. As an alternative, immersive virtual reality (VR) offers a means to simulate these experiences, referred to as “virtual forest bathing.” A key principle of forest bathing, however, is in its *multisensory* immersion [12]. While current VR applications often focus on audio-visual stimuli, recent efforts aim to enhance immersion by incorporating additional sensory elements, such as smell and haptics. These multisensory approaches promise to deepen the relaxing and restorative effects of virtual forest bathing experiences.

Some studies have already demonstrated the effectiveness of virtual forest bathing experiences in promoting relaxation and on other user experience metrics, such as immersion, presence, engagement, and flow [18, 5]. It is believed, however, that personalization of the virtual nature scenes can further enhance outcomes by tailoring the experience to individual preferences, thereby maximizing immersion and relaxation [24]. Since VR offers the freedom to design and customize environments, it is ideal for personalization. Creating virtual environments (VE) has traditionally required specialized expertise in 3D modeling and development. The rapid advancement of generative artificial intelligence (GenAI) tools, however, has opened exciting new possibilities, enabling users to create VEs via simple textual prompts.

Notwithstanding, ensuring high-quality GenAI outputs suitable for VR applications remains a challenge. This is particularly important since realism and interactivity with the environment are important factors for immersiveness and, consequently, relaxation [17]. In this short paper, we describe an ongoing project exploring the use of GenAI for personalized VE development for stress reduction (see Fig. 1 and Fig. 2). The main motivation is that personalization has been shown to maximize the sense of immersion and relaxation [25]. In our previous experiments, this was important for multisensory immersive experiences, where the congruence between the sensory elements and the VE can profoundly impact the user’s experience and relaxation states, either positively or negatively [19].

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Figure 3: In-house developed instrumented head-mounted display and electrodes and their respective modalities.

2 MULTISENSORY IMMERSIVE EXPERIENCES FOR STRESS/ ANXIETY REDUCTION

Multisensory VR experiences have been shown to promote relaxation and enhance user experience factors, such as sense of realism, presence, immersion, engagement, flow, and emotion, surpassing traditional audio-visual (AV) experiences [18, 5]. Multisensory virtual forest bathing experiences, for example, have shown to result in changes in physiological signals, such as heart rate variability, electrodermal responses, and even neural responses measured by electroencephalography (EEG) [20, 18, 5, 6, 19].

Today, several devices are emerging to integrate olfactory and other sensory elements within virtual environments (VEs) [20]. Representative examples include the SENSICS Sensory Reality Pod (SENSICS, Netherlands)¹, which combines visual, auditory, olfactory, and tactile stimuli (heat, wind, and vibrations) to create immersive experiences, the Olorama Scent Generator (Olorama Technology Ltd, Spain)², capable of diffusing up to 10 realistic scents dynamically, and the OVR ION2 (OVR Technologies, USA)³, which delivers up to nine different scents directly to the user's nose, via a scent diffuser attached to the VR headset. These devices represent a select few that integrate scent (and haptic) technologies with VEs to elevate immersion, synchronizing specific scents, tactile sensations, and temperature variations with visual and auditory content. These devices have been used to support different applications, e.g., therapy, entertainment, and relaxation [22], each offering unique capabilities and configurations.

The work in [6] showed that multisensory nature exposure could improve symptoms of post-traumatic stress disorder, which persisted even after a 3-month follow-up, as well as enhanced cognitive functions, particularly sustained attention and processing speed. The 12-session intervention also led to measurable changes in neural signatures, measured via an instrumented VR headset (see Fig. 3). Participants reported the experience to be highly immersive and had minimal cybersickness effects, highlighting its safety and feasibility for clinical applications. In the study, participants were presented with 360° videos of parks available in their city, thus were familiar to them. Conventional smells, such as pine and grass, were presented with an OVR ION 2 scent diffusion device.

The work in [19], in turn, utilized three pre-selected 360° videos of a beach, a lake, and a waterfall. The videos were scent enabled with corresponding smells and participants were given the option of choosing one video that would be presented in two conditions (audio-visual, AV, or audio-visual-olfactory, AVO) counterbalanced across subjects to avoid ordering effects. It was found that the AVO

condition showed the greatest improvement in relaxation and also a significant shift in the frontal alpha band asymmetry, supporting the withdrawal/approach behaviors often reported in the literature [35, 36, 1]. Interviews with participants post-experiment, however, showed the importance of personalization and how the congruence of the VE with the multisensory elements can be crucial to the experience. This became particularly true for the beach scenario. The video was of a Nordic beach; however, when participants were told they could choose a beach scene, some imagined they would be immersed in tropical beaches, with white sands and coconut trees, and were expecting smells of suntan/sunblock lotion and salt. This disconnect caused a break in their experience and actually affected their relaxation. One participant stated that “The smell did not agree with the reality of their beach scene. It should be more salty and sandy, not very sweet as this one was.”

3 GENAI FOR PERSONALIZED VE DEVELOPMENT: OPPORTUNITIES AND CHALLENGES

The central hypothesis driving our ongoing study is that personalization can enhance the therapeutic efficacy of VR for stress reduction, as it has shown to enhance comfort levels [30], improve user experience and promote well-being [13], deepen immersion, and promote greater relaxation [25]. Personalization may involve adapting the environment to reflect the user's preferences for scenery (e.g., forests, oceans, mountains), adjusting lighting conditions, selecting preferred soundscapes (e.g., birdsong, gentle wind, flowing water), and in the case of multisensory experiences, even selecting preferred smells and/or haptic stimuli (e.g., wind). While we have seen that standard VEs are effective at providing relaxation and fostering a sense of presence, personalization may offer the potential to further enhance these effects by aligning the VE with an individual's specific preferences, familiarity, and emotional needs [10].

In this project, we have been exploring the potential of using generative AI tools to create customized VEs “on the spot” based on the user's preferences by utilizing text prompts [33]. The generation of synthetic 3D VEs represents a significant advancement in immersive technologies, with tools leveraging generative AI to create detailed landscapes. Tools such as Blockade Labs offers a sky-box generator designed to create 360° panoramic images [3], ranging from tropical beaches to log cabins in the winter (see Fig. 1 and Fig. 2), based on text prompts. This comes with many challenges, however. For example, current GenAI tools are effective at producing only static 360° visuals. Video generation is still confined to 2-dimensional videos, which are not as immersive and may affect the overall experience. Runway's Gen-2 [29] model facilitates text-to-video synthesis, enabling users to create videos using only text prompts or images. Artbreeder [2] allows users to blend images to create unique visuals, while Deep Dream Generator [7] produces dream-like, surreal images. NVIDIA's GauGAN [23] excels in creating photorealistic landscapes, and OpenAI's CLIP-based [28] frameworks align generated visuals with user intent through natural language guidance. These tools can be adapted for specific relaxation scenarios, enhancing user engagement and well-being.

Despite their potential, these tools face limitations that need to be addressed for use in VR environments. Many generative AI systems excel at producing static images but face challenges in creating coherent, high-quality animations or dynamic scenes necessary for fully interactive VR. Moreover, the latency involved in generating complex visuals further impedes their application in real-time scenario development [21]. For instance, while tools like DALL-E and GauGAN deliver exceptional quality, their outputs are typically not optimized for rapid scene adaptation, which is crucial for personalization driven by live feedback. In terms of video generation, products such as RunwayML offer advancements in generating short sequences, typically less than 40 seconds, but challenges remain in maintaining temporal coherence and responsiveness needed for

¹<https://www.sensiks.com/>

²<https://www.olorama.com/>

³<https://ovrtechnology.com/>

continuous VR interactions [9]. Videos of such short duration may also have limited applicability for relaxation, as longer durations around 3-5 minutes are typically expected. Playing back a 40-seconds video multiple times breaks immersion and has a detrimental effect on the experience, thus more advanced methods are still needed before fully GenAI-generated 360° videos can be used.

In addition to visuals, the generation of personalized audio and smell presents other unique opportunities to enhance the immersion of virtual environments. Tools, such as Lalal's AI Sound Generator⁴ and AISFX⁵ leverage GenAI to create realistic nature sounds directly from text prompts, including wind, waterfall, wave, and bird call sounds, making them particularly suitable for immersive multisensory applications. However, challenges remain in achieving seamless synchronization of these generated soundscapes with other sensory elements, such as visual and haptic components, to create fully integrated and naturalistic virtual experiences.

Similarly, devices such as the OVR ION2, SENSIKS Pod, and Olorama rely on pre-defined scents, which can limit the freedom that users have during VE personalization. Notwithstanding, devices are emerging, such as Osmo's⁶ AI-driven scent platform, that promise the use of AI to digitize smells. Challenges with temporal synchronization of the developed smells with the VEs will need to be overcome before fully dynamic and personalized multisensory VR environments can be developed.

In addition to these abovementioned technical challenges, future studies will also have to consider non-technical challenges, including user acceptability, security, safety, and ethical concerns. For example, many individuals can be hypersensitive/allergic to certain smells and/or heat. Moreover, numerous works have highlighted how behavior and biometrics can be used to identify users in VR [4]. Personalized VEs may result in very unique biometric signatures. As such, transparency in the collection, storage, and usage of user data will be crucial for building trust and encouraging adoption of multisensory immersive experiences.

4 INVESTIGATING OPTIONS TO ADDRESS GENERATIVE AI LIMITATIONS – ONGOING WORK

Given the challenges associated with generative AI algorithms, several approaches can be considered to address their limitations while maximizing their potential for creating effective VR relaxation environments with the technology that we have today. Static image generative AI solutions, for example, where pre-rendered scenes are produced in advance based on user preferences, provide one viable option. This approach eliminates latency concerns, as the computationally intensive generative process occurs outside the user's active session. However, static scenes may lack the adaptability required to respond to real-time physiological or behavioral feedback, limiting their potential for personalization.

We are conducting a semi-personalized exploratory study inspired by the forest bathing concept, utilizing a multisensory VR experience. The experiment compares a standard scenario with a semi-personalized AI-generated scenario created using Blockade Labs. Both subjective metrics and physiological data, including EEG, electrocardiogram (ECG), and electrooculograms (EOG), are collected in real-time using an in-house instrumented VR headset (see Fig. 3). To enhance sensory stimulation, the SensiKS SDK for Unity integrates the pod with the VE, enabling real-time synchronization with multisensory elements. The experiment consists of two phases. In the first phase, participants complete a demographic questionnaire (see Supplemental Materials section) for more details) and describe their ideal relaxation scenario using text prompts. Examples of prompts and outputs are provided to guide

⁴<https://lalals.com/ai-sound-gen>

⁵<https://aisfx.org>

⁶<https://www.osmo.ai>

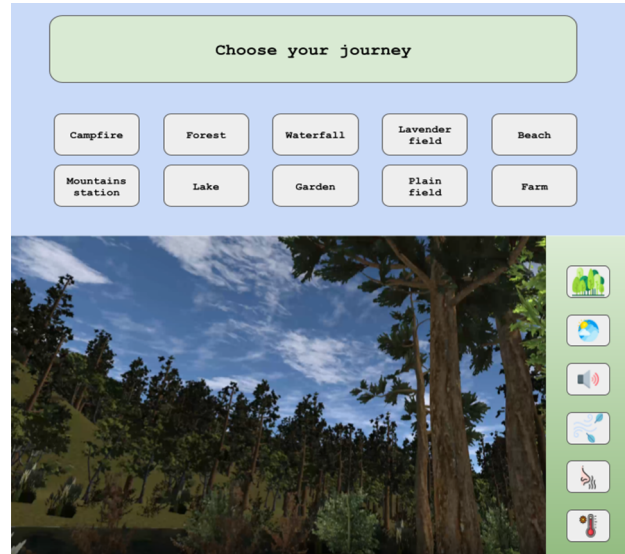


Figure 4: A screenshot of the Unity-based multisensory customization application to add interactive elements to GenAI VEs.

participants in detailing their environment. These prompts are then used to generate a skybox via Blockade Labs.

In the second phase, participants are exposed to two multisensory VR scenarios: the AI-generated skybox and a standard 360° VE, which will be selected based only on environment type (e.g., beach, forest, mountain) to provide a consistent baseline for comparison. Scenario ordering is counterbalanced to minimize ordering effects. Before each exposure, participants complete questionnaires to assess mental state, relaxation, stress, and user experience (e.g., STAI-6; International PANAS Short Form; Self-Assessment Manikin; Visual Analogue Scale for relaxation, stress, presence, immersion, fatigue, motion sickness; short version of the User Experience Questionnaire - see Supplemental Materials section). A 1-minute baseline period with a gray screen and no stimuli will be recorded prior to each condition.

Although generative AI models are not yet capable of producing 360° videos from text prompts, given the speed in which innovations are happening, they may be possible months from now. In the meantime, we have started exploring the effects of customization on relaxation potential with the tools available today. To achieve this, we are conducting a second study which allows participants to further customize their GenAI VE by (1) adding virtual elements such as trees, bushes, rocks, flowers, and campfires, (2) GenAI created audio content (e.g., from voices created with ElevenLabs to emotional music content created by AIVA to nature sounds created by AISFX and Epidemic Sound), and (3) inclusion of other multisensory elements, such as wind, heat, and aromas, via a Unity-based application, as shown in Fig. 4. Comparisons with a control group, who will not be given access to these customization abilities, will be made. This hybrid approach ensures that environments remain responsive and adaptable to user preferences while introducing dynamic movement to complement the static skybox imagery, creating a more immersive and interactive experience.

5 CONCLUSION

In this short paper, we describe two ongoing experiments designed to evaluate whether customized multisensory virtual reality experiences, using generative AI tools, can effectively increase the user's sense of relaxation and improve their experience. By combining generative AI tools with multisensory customization, we aim to

improve VR-based interventions for mental well-being via user-centred experiences. Here, we list the major challenges (technical and non-technical) with personalization of multisensory experiences using GenAI, and describe how we are addressing them in our ongoing work.

SUPPLEMENTAL MATERIALS

All questionnaires used in the experiments are available as supplemental materials and available at <https://doi.org/10.17605/OSF.IO/YJZ2B>, released under a CC BY 4.0 license.

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